Du-Du-Dudu · Formula 1 Analytics Platform

COMPLETE END-TO-END SYSTEM ARCHITECTURE, FORMULA 1 DOMAIN CONCEPTS & ANALYTICAL METHODOLOGY

1. Executive Summary

Du-Du-Dudu is a production-grade Formula 1 data engineering and race analytics platform. It ingests the complete 2025 FIA Formula One World Championship season (**24 Grand Prix, 60 sessions, 34,000+ laps**) and transforms raw telemetry, transponder timing, and race control broadcasts into high-speed analytical dashboards. Designed from the ground up to solve the challenges of real-time sports telemetry, the system combines an idempotent multi-API ETL pipeline, a PostgreSQL analytical warehouse with 7 materialized views, and an interactive Streamlit user interface.

24 / 24 Championship Rounds	**34,407** Laps Processed	**528,247** 10m Telemetry Points	**963** Pit Stops Tracked	**< 200 ms** Aggregate Query Latency

2. Formula 1 Domain Concepts Made Simple

To evaluate race engineering decisions, analysts must look beyond the finishing order. This section breaks down the foundational racing concepts modeled throughout the platform:

A. What is an Undercut Pit Stop?

The Undercut Strategy Explained

In modern Formula 1, newer tyres produce significantly more grip than old tyres—often making a car **1.0 to 2.5 seconds faster per lap**.

• **The Trigger:** Car B is stuck behind Car A. Because dirty aerodynamic turbulence ruins tyres, Car B cannot overtake on track.
• **The Move:** Car B pits **1 to 3 laps earlier** than Car A for brand-new rubber.
• **The Attack:** On its out-lap, Car B pushes at maximum speed on fresh tyres while Car A is still struggling on worn, slower tyres.
• **The Overtake:** When Car A finally pits on the next lap, its pit stop delay allows Car B to sweep past as Car A rejoins the track.

Du-Du-Dudu Implementation: The system detects all opposing driver pairs within 3.0 seconds on track that pit within 3 laps of each other, verifying whether the earlier pitter gained track position without the influence of Safety Cars (labeled **ESTIMATED**).

B. Pit Stop Stationary Time vs. Total Pit Loss Delta

Fans usually see the TV graphic showing a **2.2-second wheel-change**, but the strategic cost of a pit stop is much larger. Cars must decelerate to the pit lane speed limit (typically 80 km/h or 60 km/h in Monaco) across several hundred meters of asphalt while competitors fly down the straight at 320 km/h.

• **Pit Stop Duration:** The exact time the car is stationary in the pit box having tyres bolted on (transponder raw time).
• **Pit Loss Delta:** The total racing time lost relative to a standard flying lap: (In-Lap Time + Out-Lap Time) - (2 × Driver Median Clean Flying Lap). Depending on the circuit, this ranges from **18.5s** (Red Bull Ring) to over **26.5s** (Silverstone / Singapore).

C. What is Tyre Degradation & Stint Progression?

Tyres are the single biggest performance variable in Formula 1. Pirelli supplies three dry-weather slick compounds to each race: **Soft (C5/C4/C3)** (highest initial grip, fast degradation), **Medium**, and **Hard** (durable, steady pace). As a tyre undergoes thermal cycles and tread loss, lap times increase linearly until reaching the 'cliff' where grip drops catastrophically. **Du-Du-Dudu** tracks exact tyre life per lap (1, 2, 3... N) across stints, plotting lap time delta versus tyre age to isolate the degradation coefficient (seconds lost per lap).

D. What is a 'Clean Lap' and Why Filter Them?

Raw lap times are contaminated by external chaos: opening lap traffic, Safety Cars (SC), Virtual Safety Cars (VSC), red flags, and in/out laps. To measure a car's genuine pace, **Du-Du-Dudu** applies a strict deterministic 5-point cleaning rule:
1. **Lap 1 Excluded:** Standing starts distort pace.
2. **No Pit In/Out Laps:** Speed limiters artificially inflate lap times.
3. **Green Flag Only:** Track status must be strictly Green (`1`). All SC, VSC, and Yellow flag laps are removed.
4. **Timing Accuracy:** Transponder `is_accurate` flag must be true.
5. **Dry Compounds Only:** Wet and Intermediate laps are separated from dry slick baselines.

E. Spatial Telemetry & 10-Meter Grid Resampling

F1 cars log telemetry against elapsed time. However, comparing two drivers across time is flawed because Driver A might reach Turn 3 in 24.1s while Driver B reaches it in 24.4s. To perform a mathematically sound comparison, **Du-Du-Dudu resamples all telemetry onto a normalized spatial grid spaced at exact 10-meter intervals** (0m, 10m, 20m ... 5,280m). This allows instant overlay of speed, throttle, brake pressure, and gear changes at the exact same physical coordinates on the track.

3. End-to-End System Architecture

The architecture is separated into three decoupled tiers: an **Ingestion Engine**, a **PostgreSQL Analytics Warehouse**, and an **Analytical Presentation Layer**.

Layer	Technology	Role & Key Engineering Responsibilities
Ingestion & ETL	Python, FastF1, Jolpica-F1, OpenF1, Pandas	Tri-API orchestrator with automated fallback. Fetches sessions, results, 34K+ laps, pit stops, race control broadcasts, and 10m spatial telemetry.
Storage & Aggregates	PostgreSQL 16 / Supabase, SQLAlchemy, psycopg 3	Normalized relational schema with 7 materialized views and unique indexes. Features connection pooling (`pool_pre_ping`, PgBouncer bypass threshold).
Presentation & Visuals	Streamlit, Plotly, Custom Dark F1 CSS	Zero raw SQL queries; all dashboard views pull directly from indexed materialized views for sub-second page loads across all 24 Grand Prix.

4. The Tri-API Ingestion Pipeline & Fallback Strategy

Official F1 data feeds frequently restrict live timing CDN caches with HTTP 401 Unauthorized errors for uncached sessions. To ensure robust data availability across all 24 events, **Du-Du-Dudu** implements a resilient multi-tier fallback architecture:

- **FastF1 API:** Primary high-resolution source for local cached timing, sector splits, speed traps, and car telemetry traces.
- **Jolpica-F1 (Ergast API):** Automated HTTP REST fallback. When FastF1 is restricted, Jolpica seamlessly provisions full race results, qualifying grids, and transponder lap times.
- **OpenF1 Stint Enrichment:** Jolpica lap data lacks tyre compounds (marking them 'UNKNOWN'). The platform's etl/stints.py queries OpenF1's live stint endpoints, enriching **25,584 laps** with exact Soft, Medium, and Hard compounds and tyre life counters.
- **Race Control Messages:** Ingests 2,378 race control messages to map precise start and end laps for Safety Cars and Virtual Safety Cars.

5. Database Warehouse & The 7 Materialized Views

To eliminate heavy SQL window functions and aggregations during dashboard browsing, all heavy math is precomputed into **7 Materialized Views** with unique indexes for concurrent refresh:

Materialized View	Granularity	Analytical Metric Precomputed
mv_clean_lap_pace	Driver per session	Median clean lap time, 25th/75th percentiles (IQR), fastest clean lap, total valid clean laps.
mv_pit_loss	Driver pit stop	In-lap, out-lap, stationary duration, and net pit lane loss delta relative to flying baseline.
mv_undercut	Driver pair per stop	Gap before stop, gap after stop, net delta, and undercut success boolean flag.
mv_tyre_deg	Compound & tyre age	Average and median pace delta from baseline as tyre age increases from lap 1 to 35+.
mv_teammate_gap	Constructor per session	Driver 1 vs Driver 2 qualifying delta (ms) and median clean race pace delta (ms).
mv_grid_vs_finish	Driver per race	Grid position, classified finish position, net positions gained or lost (+/-).
mv_reliability	Constructor per season	Total starts, finishes, mechanical retirements (DNF), crash DNFs, and completion rate %.

6. Dashboard Modules & Visualizations

The web interface is organized into four intuitive analytical modules built with Streamlit and Plotly:

- **Module 1: Race Analysis:** Complete Grand Prix classification, Grid vs Finish position delta bar charts, stint tyre compound timelines, and semi-transparent Safety Car/VSC overlays.
- **Module 2: Pace Analysis:** Clean lap pace IQR boxplots ranking car speed, compound-specific degradation slope charts (Soft vs Medium vs Hard), and intra-team qualifying/race pace differentials.
- **Module 3: Pit Stop Execution:** Pit loss delta distribution by driver and team, stationary box time rankings, and undercut encounter tables displaying gaps before and after pit windows.

• **Module 4: Driver Telemetry & Mini-Sectors:** Side-by-side fastest lap comparison on a 10m spatial grid. Features speed traces with corner apex markers, time delta trace, pedal inputs, 25-segment mini-sector track dominance, and corner braking/apex statistics.

7. Engineering Principles & Gravitas Standards

Every calculation published in **Du-Du-Dudu** adheres to strict software engineering and data provenance standards:

• **The ESTIMATED Badge Philosophy:** Transponder-recorded values (lap times, sector times, trap speeds) are presented as pure ground truth. Any derived calculation (pit loss deltas, undercut pairs, resampled telemetry, corner statistics) is explicitly marked with a visible ESTIMATED badge in the user interface to ensure transparent methodology.

• **Automated Data Quality Suite:** etl/quality.py runs validation checks on every session load—flagging missing laps, implausible lap times (<50s or >180s), and non-monotonic stint numbers.

• **Comprehensive Test Coverage:** The platform includes **73 automated unit and integration tests** covering cleaning rules, outlier detection, sprint round logic, and telemetry math (100% test pass rate).